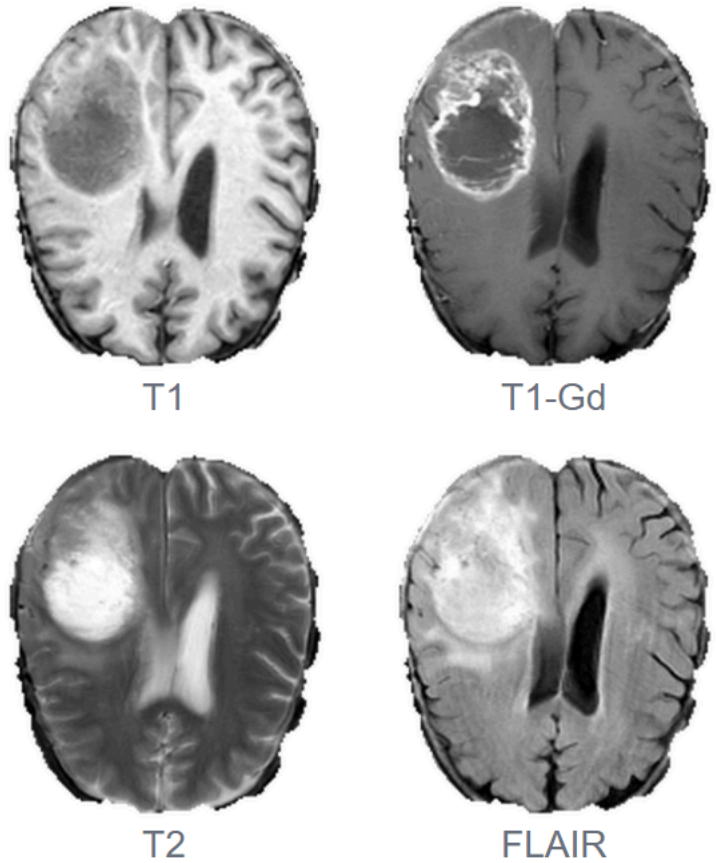
An exploratory multi-scale analysis linking MRI radiomics with *KDR*/VEGFR-2 expression and kinase-domain structure in glioblastoma

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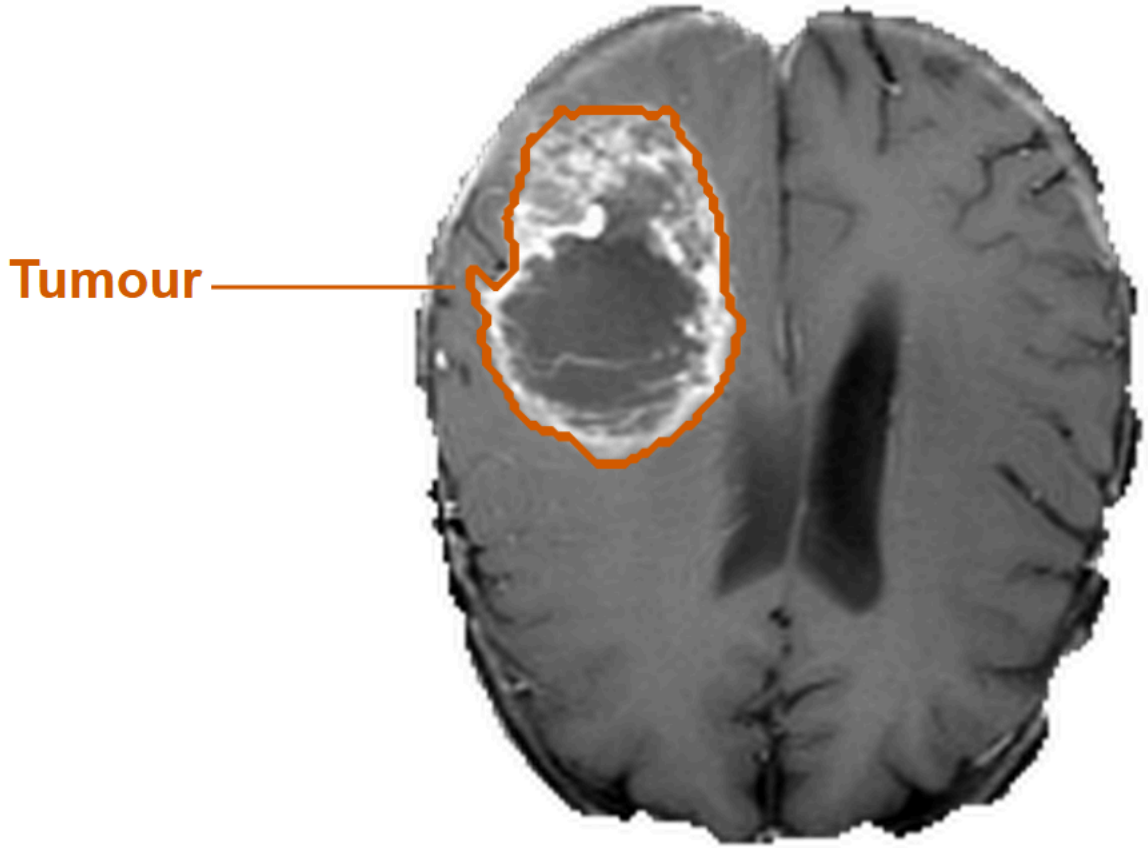


Glioblastoma has a median survival of about 15 months under standard care

15

months, median survival

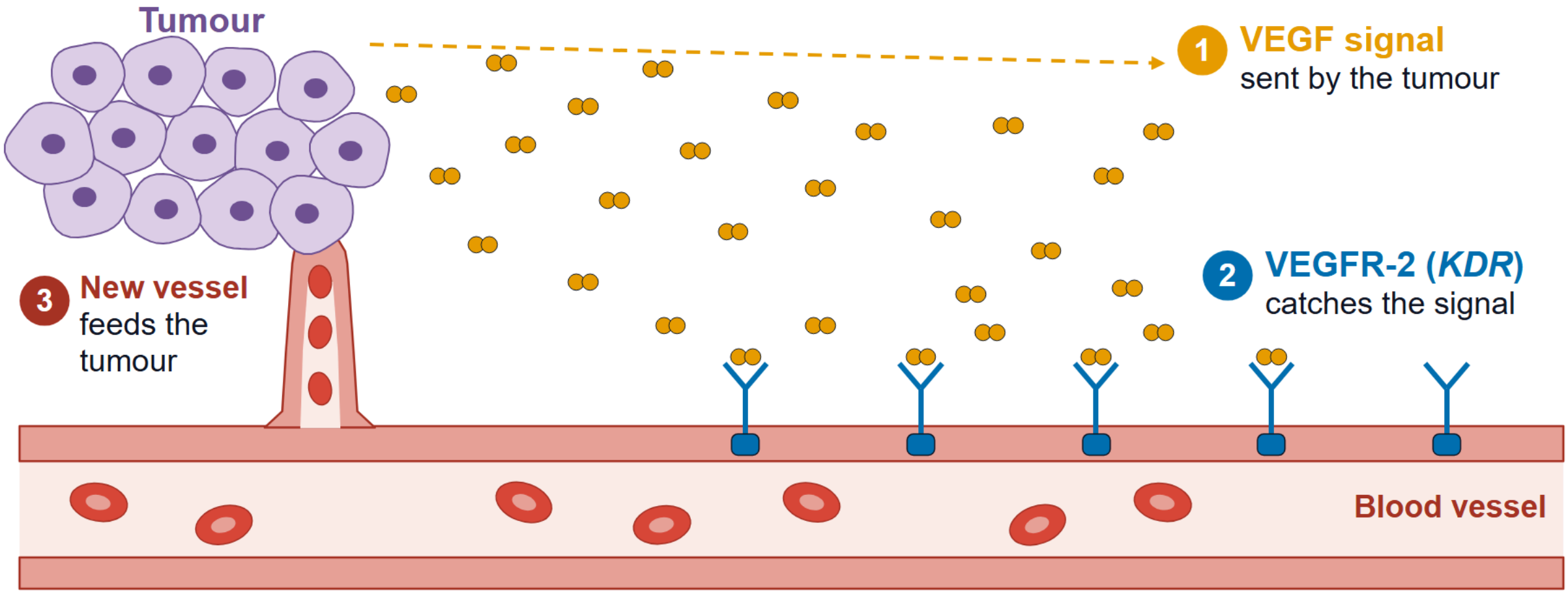
Glioblastoma is the most aggressive primary brain tumour



Real patient MRI (contrast T1), public TCGA-GBM data

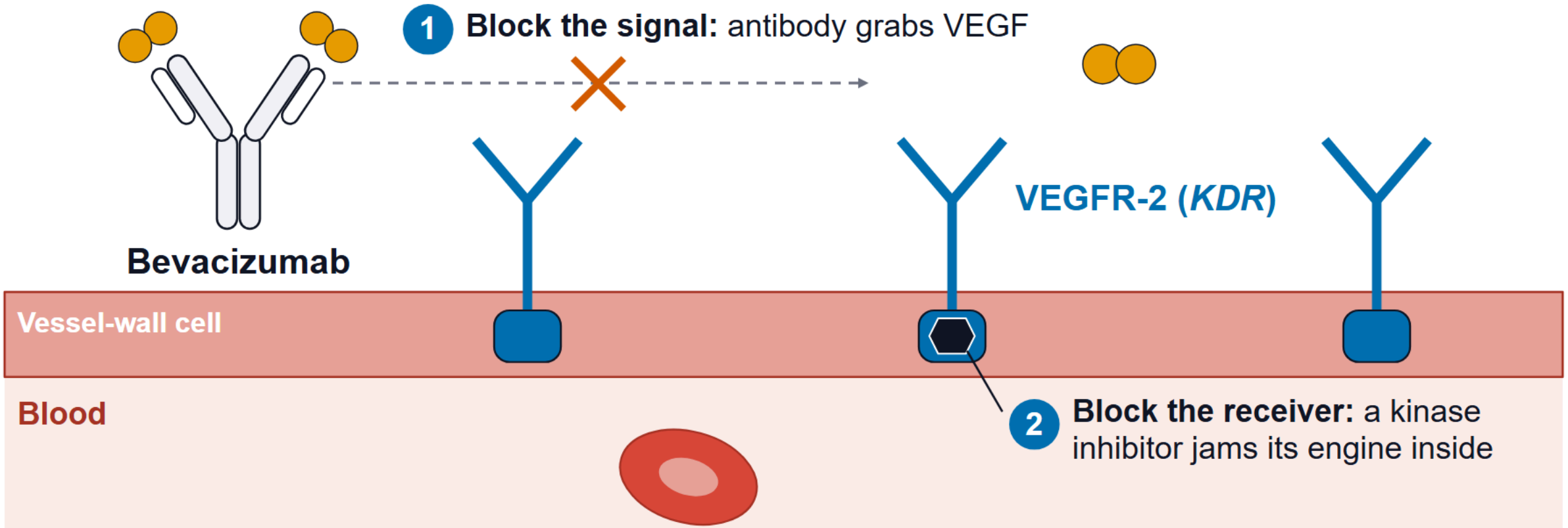
Stupp et al., N Engl J Med 2005

Tumours recruit new blood vessels through VEGF signalling at VEGFR-2, encoded by *KDR*



Holmes et al., Cell Signal 2007

Anti-angiogenic drugs act on this pathway

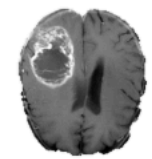


Carvalho et al., Sci Rep 2021; McTigue et al., PNAS 2012

KDR status is read from tumour tissue; can MRI provide a non-invasive alternative?

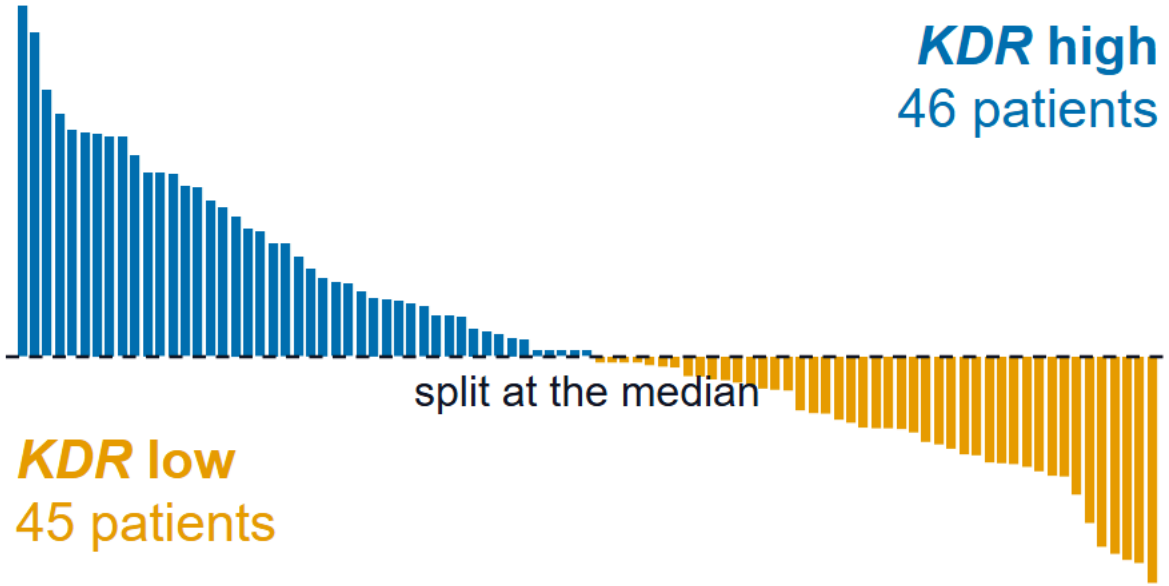
Today: tissue from surgery or biopsy

Risky, and hard to repeat as the tumour changes



Question: predict *KDR* level from MRI alone

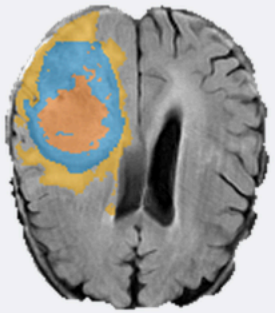
91 patients with both MRI and *KDR* data



Each bar = one patient, ranked by *KDR* level (tumour tissue)
TCGA-GBM, public data

Bakas et al., Sci Data 2017; Cerami et al., Cancer Discov 2012

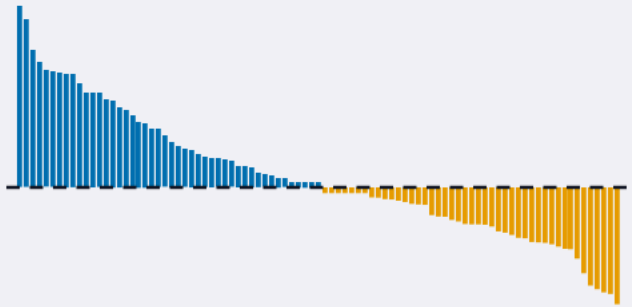
A three-scale framework: image, gene, structure



1 Scan → numbers

Software measures the tumour: size, shape, edge roughness, brightness

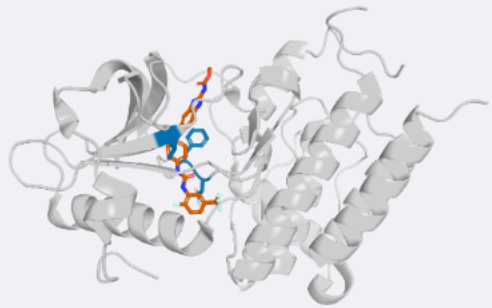
T1, T1-Gd, T2, FLAIR · PyRadiomics



2 Do the numbers follow the gene?

Which measurements rise and fall with *KDR*? Corrected for chance

A model then guesses high or low for patients it never saw



3 Where do drugs bind?

The pocket on the VEGFR-2 protein where inhibitor drugs sit

PDB 2OH4 · PyMOL

van Griethuysen et al., Cancer Res 2017; Hasegawa et al., J Med Chem 2007

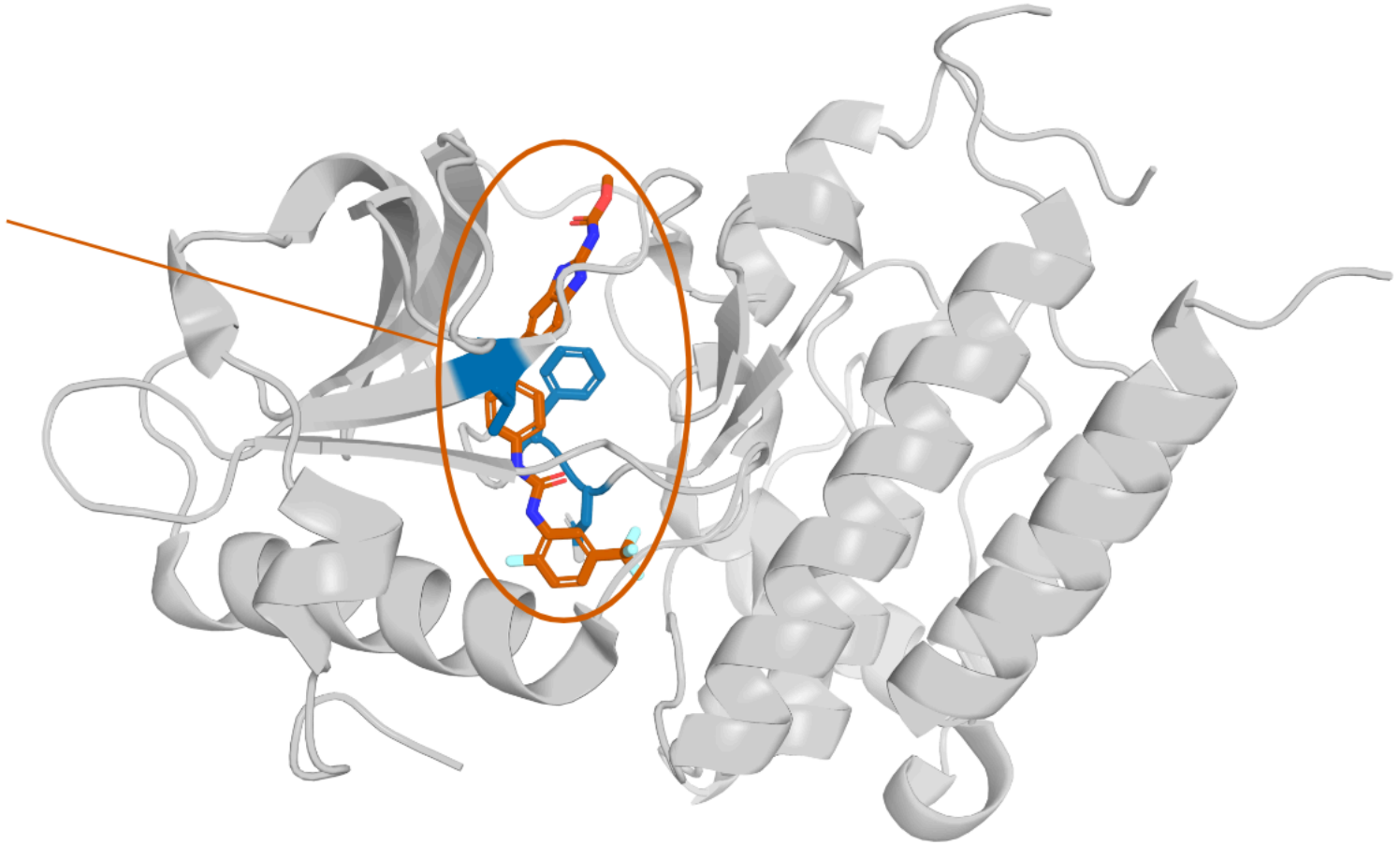
The VEGFR-2 kinase domain binds the inhibitor in its ATP pocket

The receptor's engine
kinase domain, inside the cell

A drug molecule
inhibitor, sitting in the pocket

Pocket walls
Val914, Asp1044, Phe1045

Like a key jammed in a lock: ATP, the fuel that switches the receptor on, cannot get in



PDB 2OH4 (2.05 Å), crystal structure; rendered in PyMOL

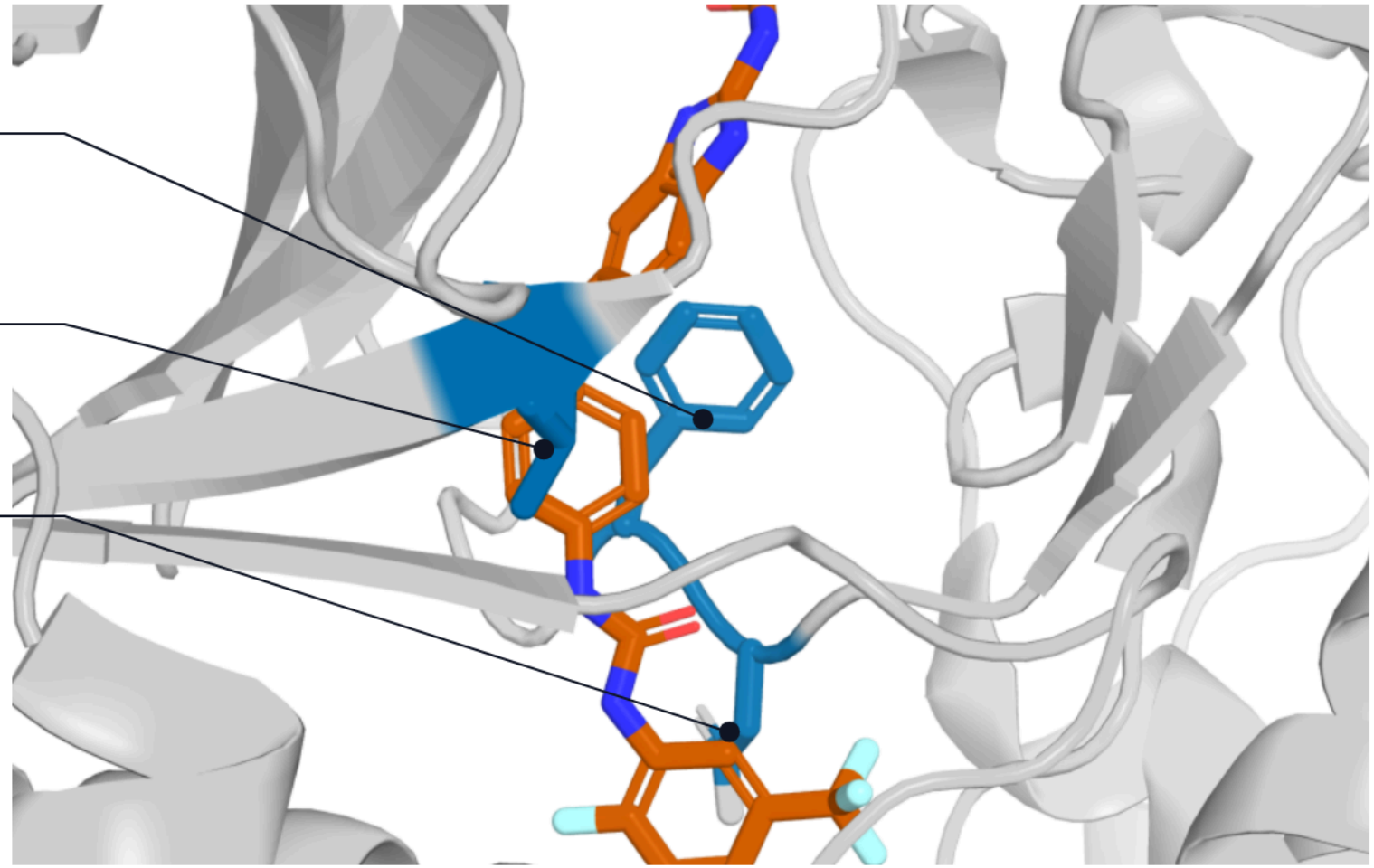
Hasegawa et al., J Med Chem 2007

Eight residues line the pocket, including the gatekeeper Val914

Phe1045 (DFG motif)

Val914 (gatekeeper)

Asp1044 (DFG motif)



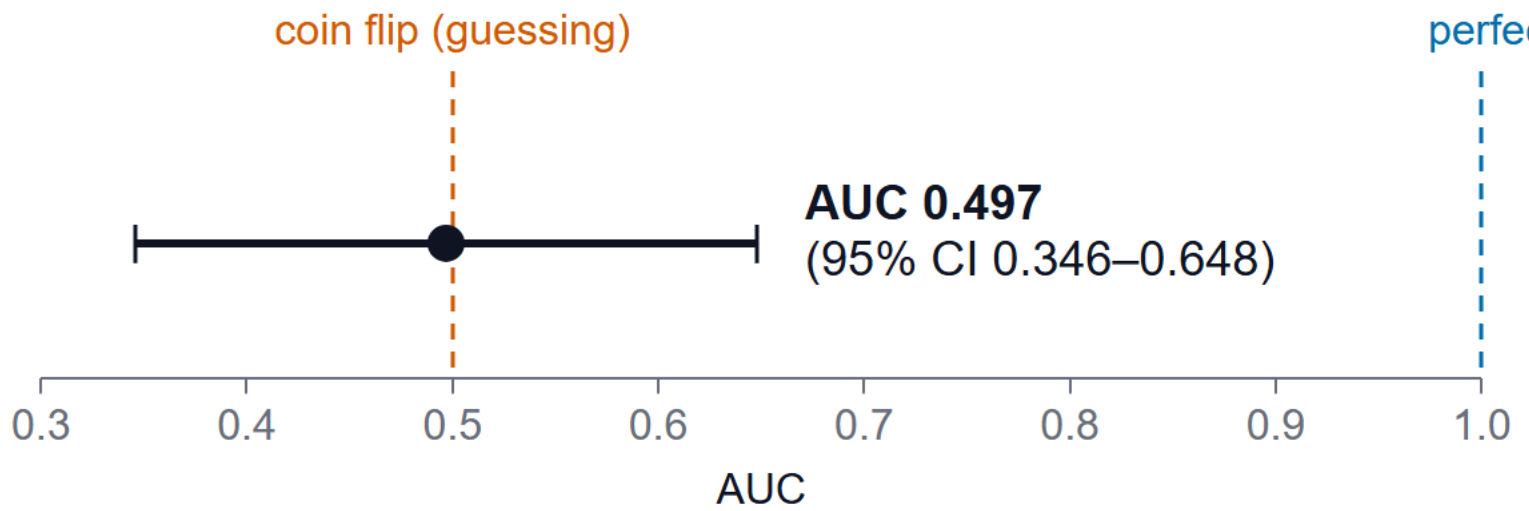
Crystal numbering;
UniProt numbering is two higher

Qualitative context, not linked to the imaging model

McTigue et al., PNAS 2012

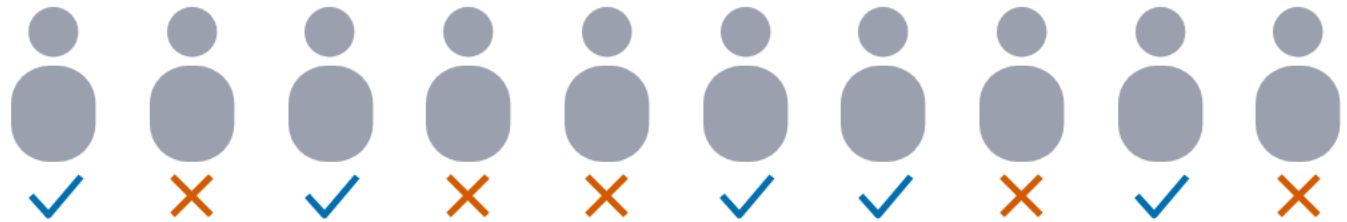
MRI radiomics did not predict *KDR* status in unseen patients

Our model
unseen patients (n = 91)



Not yet

About half of patients would be put in the wrong group



Tick = correct group · cross = wrong group

This study

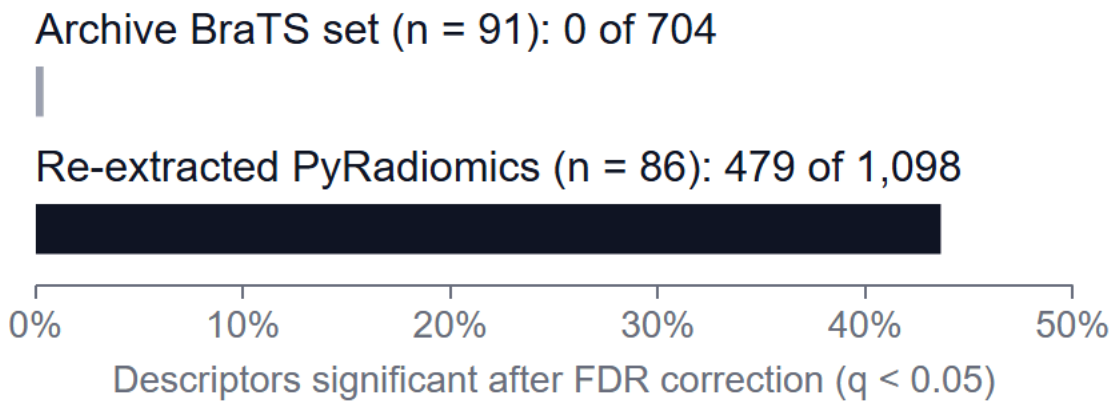
Three reasons the signal is not yet usable

Small cohort

91 patients, one cohort, no external validation, no scanner harmonisation

Results depend on feature extraction

Same images; the sets also differ in patients and definitions



Weak signal

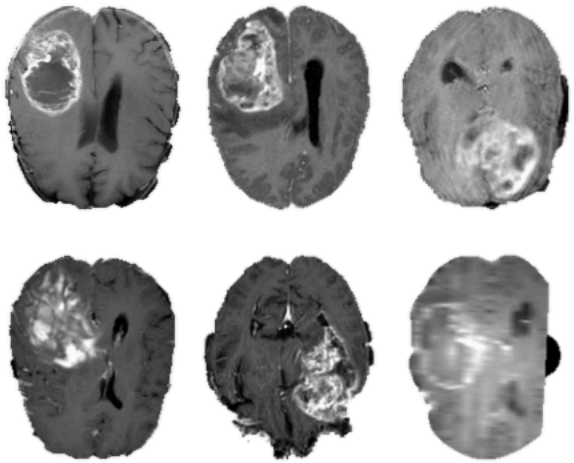
Measurable but weak association; may be mixed with other biological signals

This study

Next: larger independent cohorts and models built for weak signals

1. Replicate the FLAIR necrotic-core association in an independent multi-institutional cohort, with multiple-testing control and validation on unseen patients

2. Try models designed for weak signals in small data, and include other genes beyond *KDR*



More patients, more hospitals

For now, MRI cannot replace tissue for reading *KDR*.

This study

Thank you

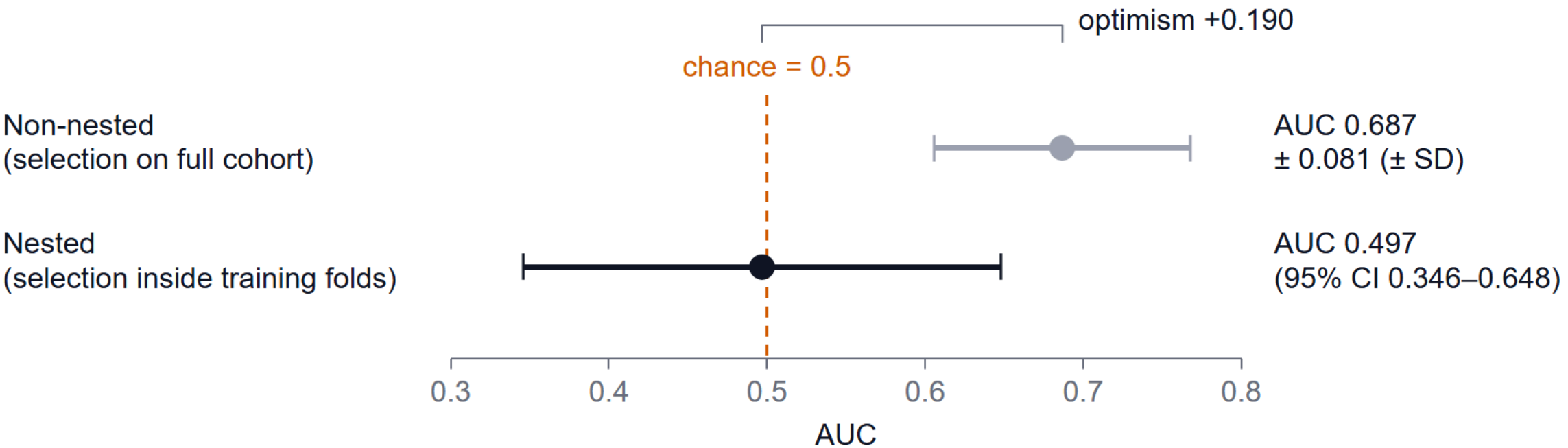
ขอบคุณครับ

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ipb.link/triu-nugraha-2026

Backup: validation design · software comparison · protein structure · references

Selecting features on the full cohort inflated the AUC by 0.190



Accuracy 48.4% (nested)
Median split z = −0.179 · 46 high / 45 low

This study

Same images, two extractions: 0 versus 479 descriptors survive FDR correction

Descriptor set	Patients	Tested	Nominal	FDR q < 0.05	Min q
Pre-computed BraTS	91	704	61	0	0.530
Re-extracted PyRadiomics	86	1,098	682	479	0.010

Expected by chance ≈ 35 of 704 (pre-computed set)

Strongest: FLAIR necrotic-core feature, ρ = 0.432, q = 0.010

479 features reflect a few correlated dimensions, not 479 independent findings

Genes (6 transcripts tested): *KDR* q = 0.010; *CXCL8* q = 0.039

This study (Table 1)

VEGFR-2 kinase domain with inhibitor (PDB 2OH4)

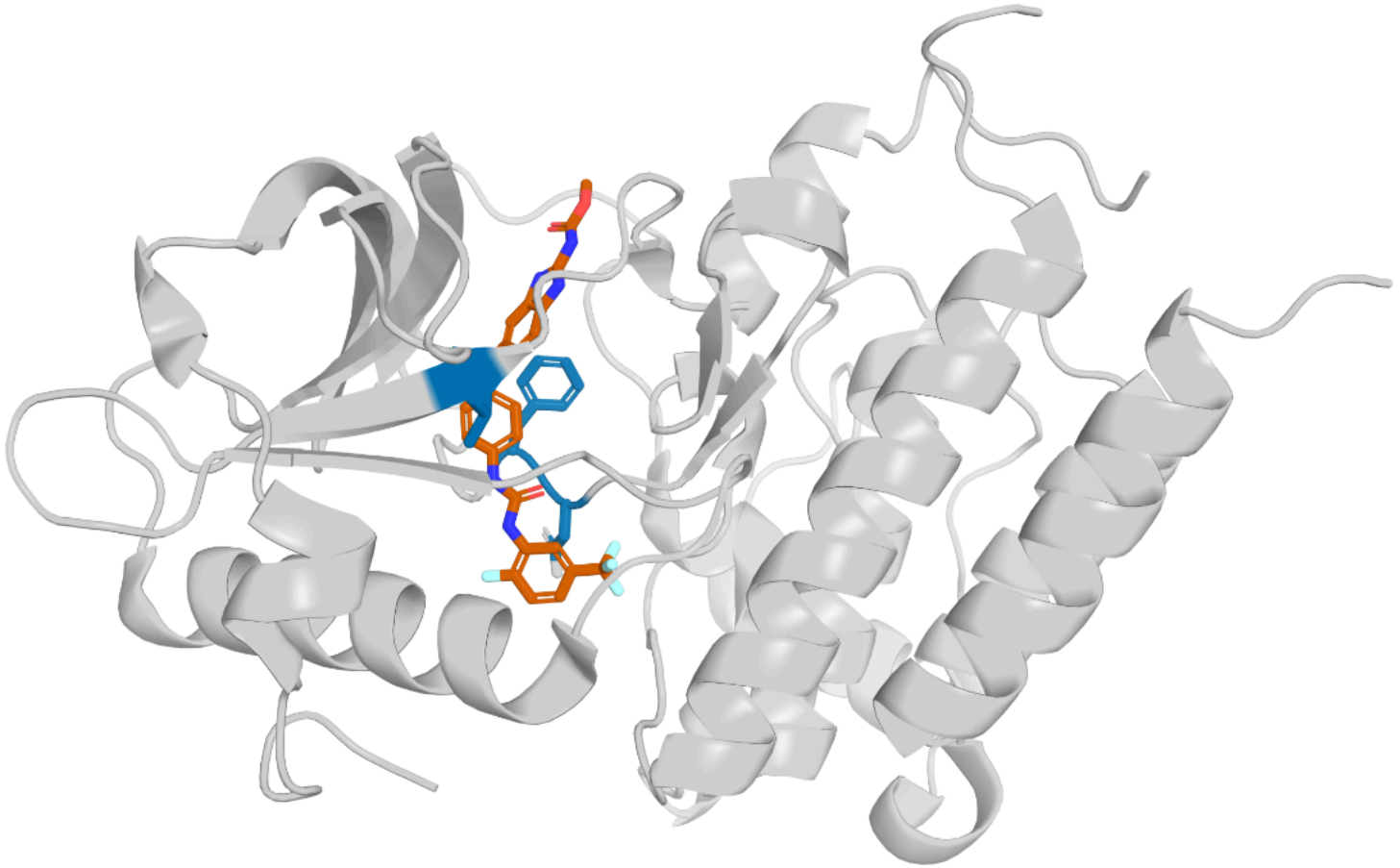
Eight ATP-binding residues, including:

Val914 (gatekeeper)
Asp1044, Phe1045 (DFG motif)

Residues within 5 Å of the inhibitor (PyMOL), cross-checked against the literature

Crystal numbering; UniProt numbering is two higher

PDB 2OH4, 2.05 Å resolution



Hasegawa et al., J Med Chem 2007; McTigue et al., PNAS 2012

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